DIY PROJECT · PHY341/745

Shock waves in a one-dimensional fluid

How much of what you see is the grid?

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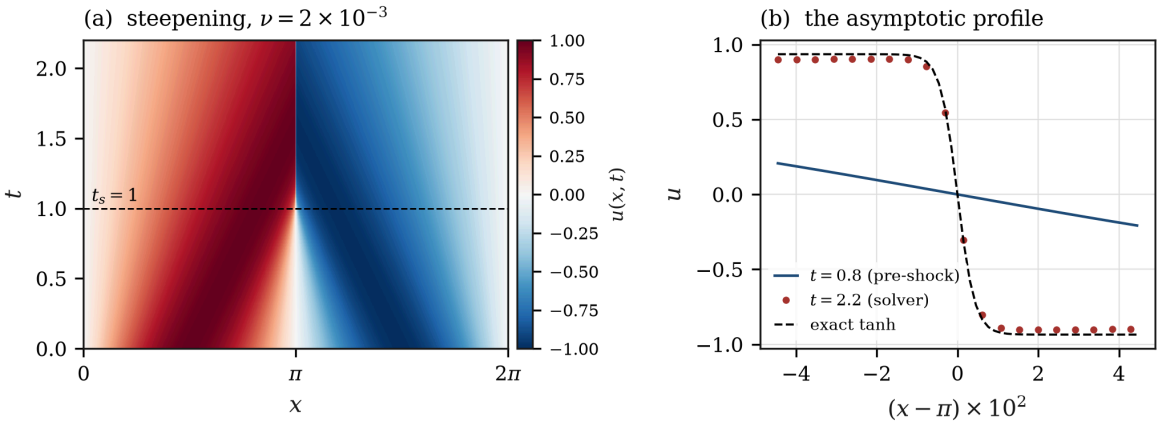
Physics Computer Lab · NISER Bhubaneswar

One equation

$$u_t + uu_x = \nu u_{xx}$$

- nonlinearity **steepens** the wave
- viscosity **smooths** it
- the balance is a **shock**

In one dimension, incompressibility forces $\partial_x u = 0$: advection and the pressure gradient both vanish. Dropping only the pressure gradient leaves the **1-D Navier–Stokes** equation.



The whole talk in one picture: steepening, breaking at $t = 1$, and the asymptotic profile.

The only theory needed

When does it break?

Characteristics $x = \xi + u_0(\xi)\,t$ cross at

$$t_s = -\frac{1}{\min} u_{0'}(x) = 1$$

How thick is what remains?

$$\delta = 4\nu/U$$

$$u = -U \tanh[U(x - x_0)/2\nu]$$

Four predictions, no free parameters

$$E(k) \propto k^{-2}$$

$$\zeta_p = \min(p, 1)$$

Everything in this talk is measured against one of these. That is what makes the *numerics* measurable rather than decorative.

Five solvers, three diagnostics

scheme	order
upwind	1
Lax–Wendroff	2
FTCS	2 (unstable)
Crank–Nicolson + Heun	2
spectral, Strang + RK4	exponential

All written for this project. NumPy only for arrays — no `np.fft`, no `scipy`. The FFT is radix-2 and checked against a naive DFT to 10^{-15}.

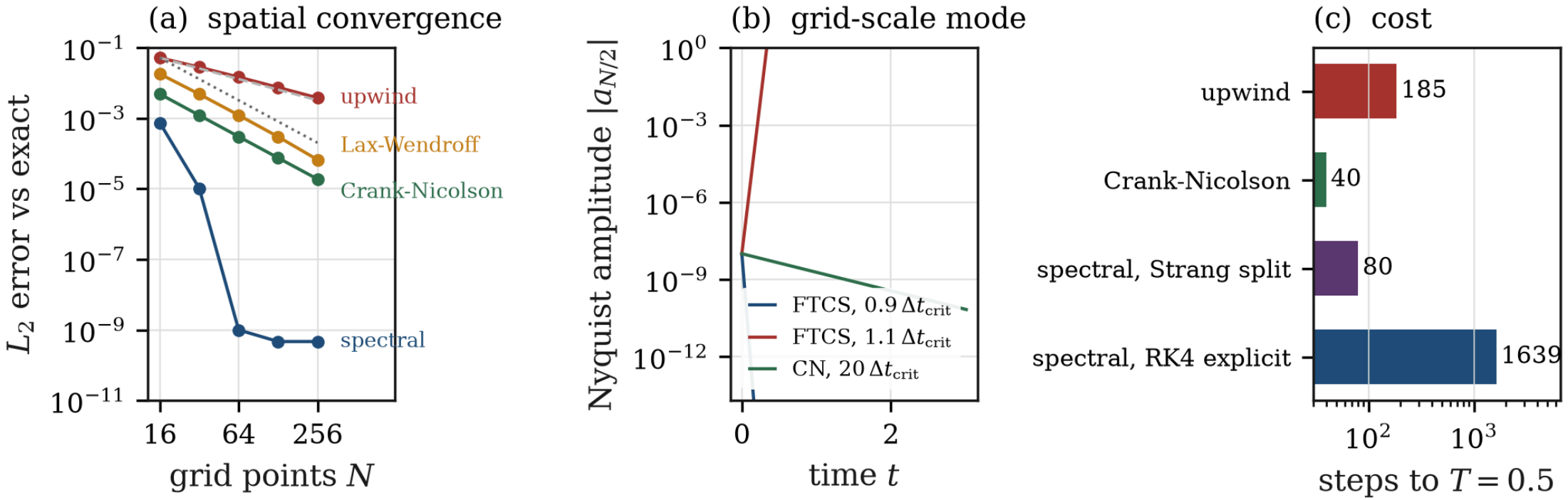
Diagnostics

$\delta =$ jump / max slope
$E(k)$ from the from-scratch FFT
$S_p(r)$, moments of increments

Reference

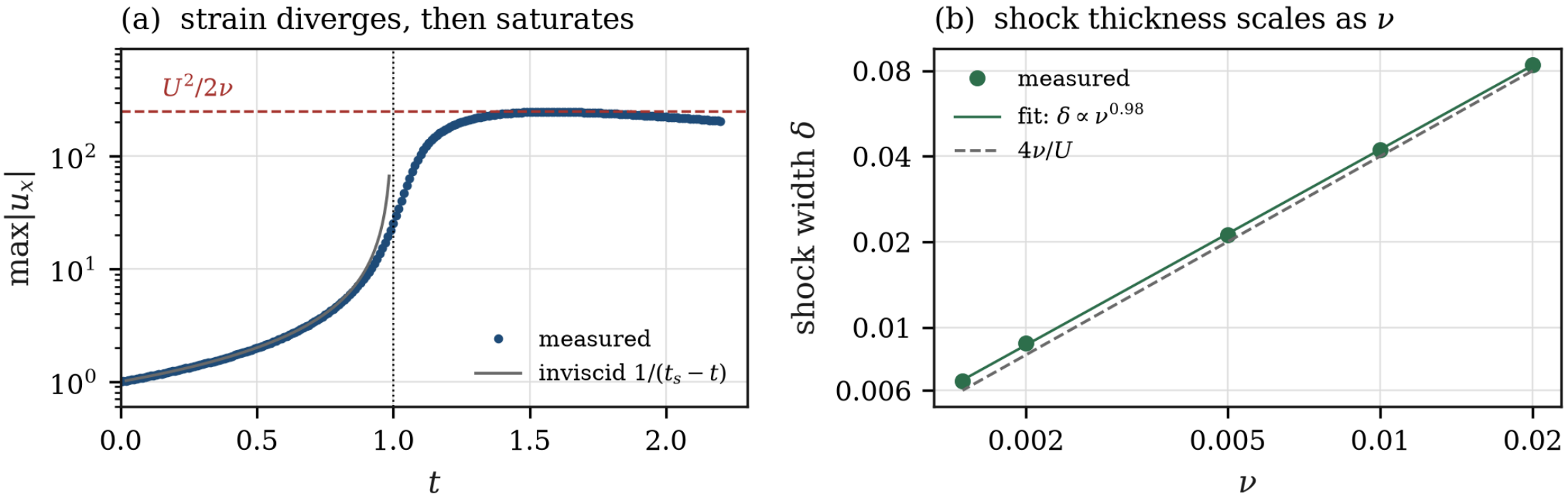
exact Cole–Hopf solution, grid-independent to 10^{-12}

Result 1 — are the solvers right?



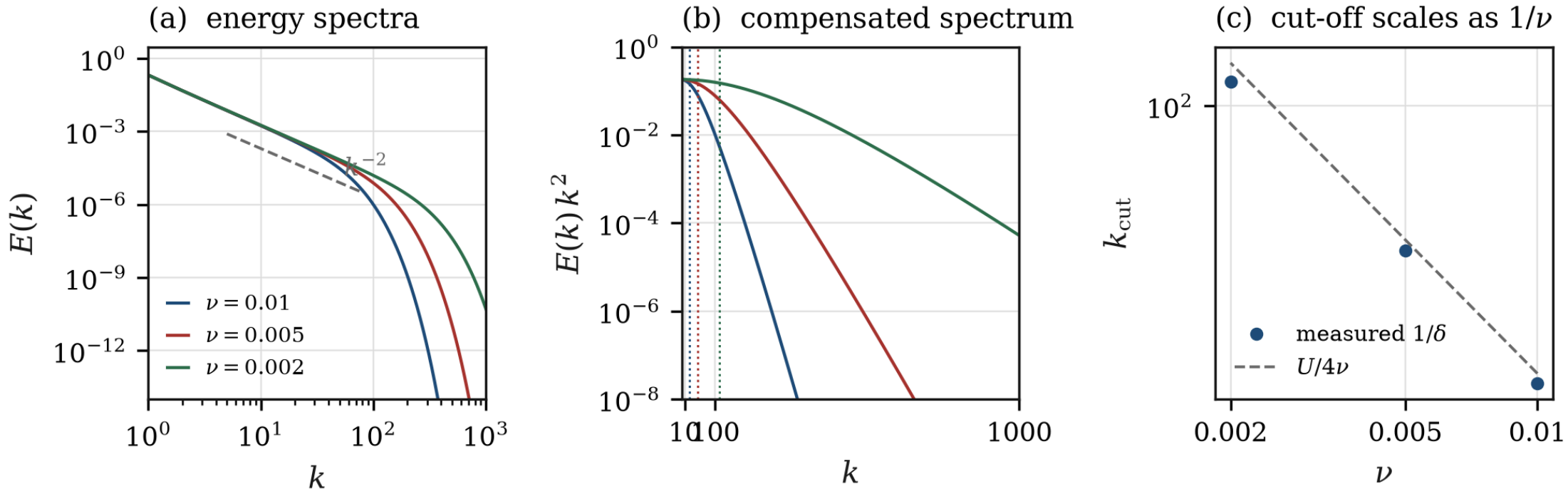
measured orders **0.94, 2.02, 2.00**, spectral $7.5 \times 10^{-4} \rightarrow 10^{-9}$; growth rates match von Neumann theory to **0.3%**.

Result 2 — breaking, and what is left behind



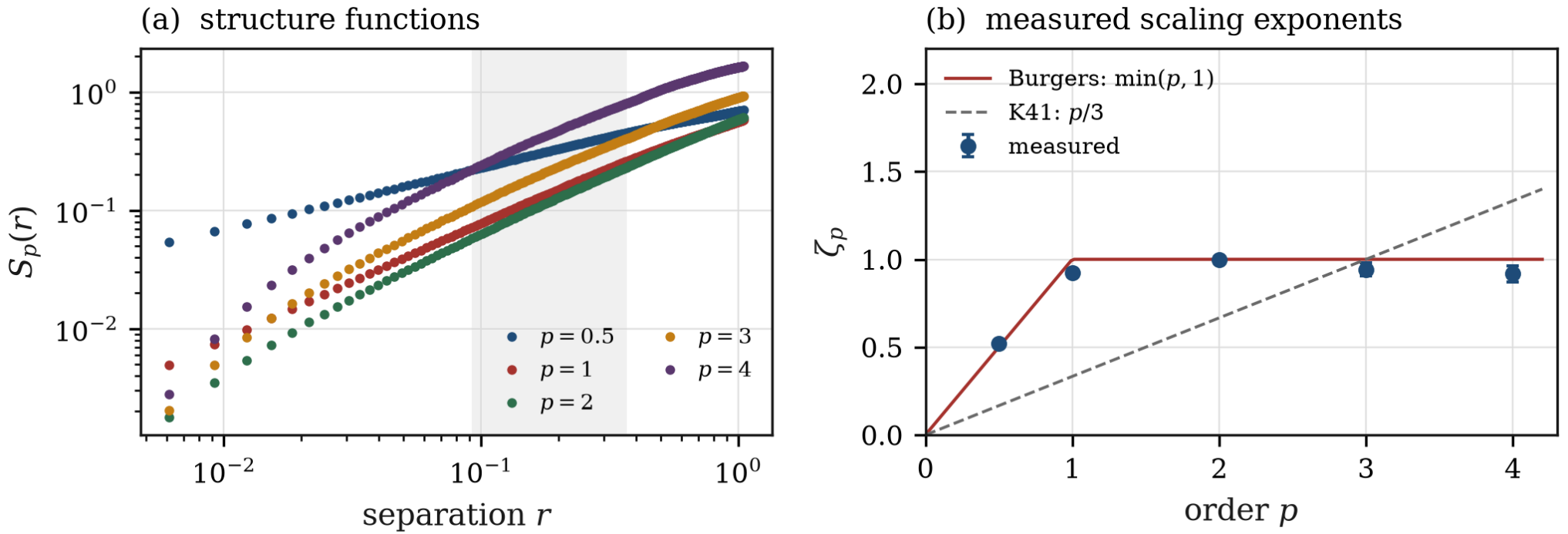
strain $= 1/(t_s - t)$ to 0.5%, saturating at 247.5 **vs** 250 $\delta \propto \nu^{0.985}$, prefactor 5–10% above $4\nu/U$.

Result 3 — the spectrum has a movable end



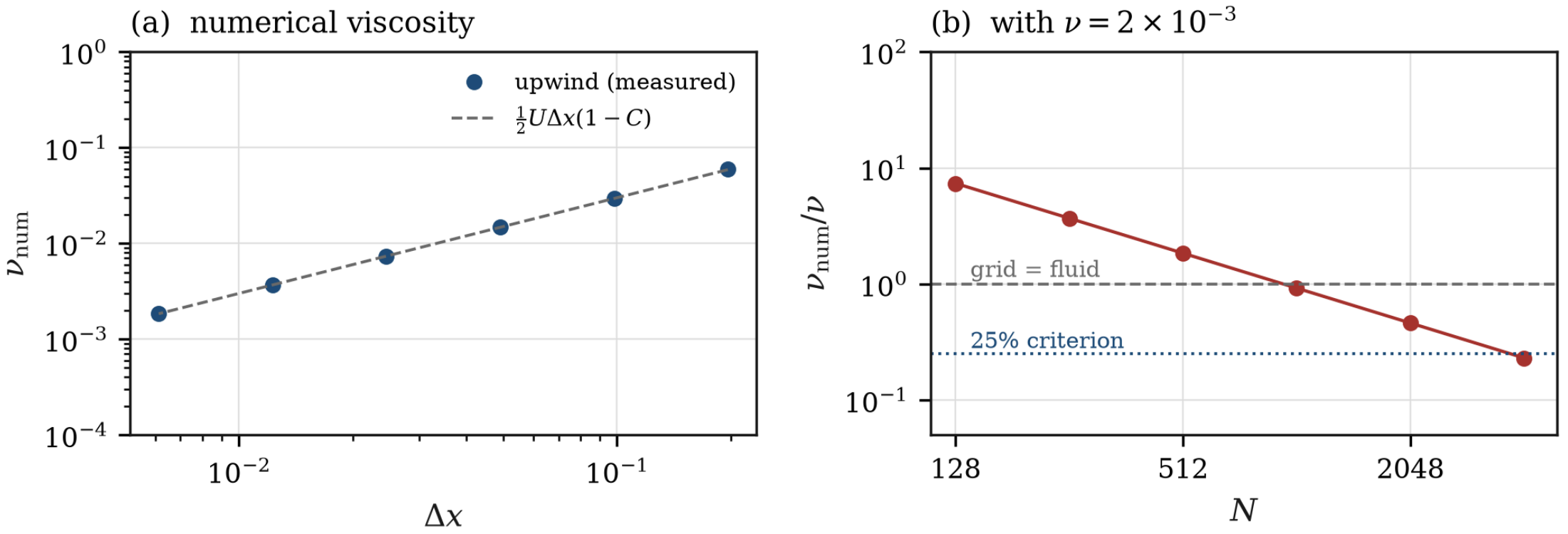
$E(k) \propto k^{-2.030 \pm 0.001}$ for every ν; the cut-off $k_{\mathrm{cut}} \propto \nu^{-0.971}$. Viscosity changes where the spectrum *ends*, not its shape.

Result 4 — many shocks, one exponent



$\zeta_p = 0.52, 0.92, 1.00, 0.94, 0.92$ versus $\min(p, 1)$ and $p/3$: a factor of three apart at $p = 4$, with a 5% error bar.

Result 5 — how much of the viscosity is the grid?



$\nu_{\mathrm{num}} = U\,\Delta x\,(1-C)/2$ verified to **1%** over five decades; at $N = 256$ the grid is 3.7 × more viscous than the fluid.

What I take away

- Convergence *order* does not predict shock *quality*: all four schemes have known orders and disagree qualitatively — smeared vs ringed.
- Integral quantities are robust, local ones are not: the k^{-2} slope survives a 10^3 change in ν; the shock thickness does not.
- Numerical dissipation is measurable, has a formula, and should be computed rather than hoped away.

> **Practical criterion.**
> Resolve the shock, not the wavelength: $\Delta x \leq 0.2\,\delta$ — at least five points per shock.

Limitations, and what I would do next

Limitations

- inertial range only $\approx$ 1.5 decades
- statistics at one time, not stationary
- one equation; no vortex stretching (absent in 1-D)

Next

- forced, statistically stationary run
- WENO/TVD scheme, same ζ_p test
- extension to 2-D

References & code

1. J. M. Burgers, *Adv. Appl. Mech.* **1**, 171 (1948).
2. J. Bec and K. Khanin, *Phys. Rep.* **447**, 1 (2007).
3. G. B. Whitham, *Linear and Nonlinear Waves* (Wiley, 1974), ch. 2–4.
4. R. J. LeVeque, *Finite Volume Methods for Hyperbolic Problems* (CUP, 2002), ch. 4, 6.
5. J. P. Boyd, *Chebyshev and Fourier Spectral Methods* (Dover, 2001), ch. 2.
6. J. D. Cole, *Quart. Appl. Math.* **9**, 225 (1951).

> Code, figures and every number:
> **github.com/Aryans-lab/Computational_Physics_Lab** → `project_burgulence/`